

Topological phonon gaps in Fibonacci isotope nanoribbons of hexagonal boron nitride

Ronghua Wu

Zhuhai Jiuchongtian Aviation Technology Co., Ltd., Zhuhai, China
wrh@gdtzn.com

Ruqing Chen

GUT Geoservice Inc., Montreal, Canada
ruqing@hotmail.com

August 24, 2026

Abstract

Hyperuniform isotope order suppresses phonon scattering through a power law that transverse channel averaging cannot disturb [R. Wu and R. Chen, Zenodo (2026)]. Here we invert the objective and ask for the opposite spectral object: a topologically labelled phonon gap, which the same averaging destroys. Survival therefore demands a strictly single-mode nanoribbon, and with it a confrontation with edge-roughness decoherence. Anchoring the roughness scattering to a grid-converged boundary-conforming Green’s-function calculation, validated against an analytic Born result, we find that the Ziman ray formula underestimates the coherence length by an order of magnitude in the single-mode wave limit $\lambda \gg W$: a zero-parameter law $\xi_{\text{edge}} = 2W^2/[q^2 S_w(2q)]$ reproduces a 15-point numerical scan with all three exponents (frequency to 10^{-3}) and magnitude to 3.5%, softening the edge-flatness requirement from atomic ($\lesssim 0.13$ nm) to lithographic ($\lesssim 1$ nm). A Fibonacci $^{10}\text{BN}/^{11}\text{BN}$ modulation of period 24.8 nm in a 30 nm ribbon places a Chern-labelled gap at 0.250 THz at a factor 2.20 cost in localization length relative to a periodic superlattice. Gap labelling is verified falsifiably (median residual 3×10^{-6} , $1385\times$ below random controls) and phason pumping exhibits the full counter-propagating edge-state structure, with per-edge spectral flow $\pm q$ for all twelve principal gaps up to $|q| = 11$. We propose an asynchronous-optical-sampling cavity measurement with a built-in opposite-Chern-number control gap.

1 Introduction

In a companion work [1] we established that the relaxation rate for phonon scattering by a spatially correlated isotope arrangement obeys $\tau_{\text{iso}}^{-1} \propto \omega^{d+1+\alpha}$, the hyperuniformity exponent α of the arrangement [2] entering as an additive, dimension-independent correction to the Rayleigh–Tamura exponent [3]. The conclusion drawn there was suppressive: ordering the isotopes *removes* long-wavelength scattering. The present work inverts the objective, and in doing so inverts every geometric conclusion that followed from it.

The mirror.—A power law is a scale invariant. Averaging it over the transverse channels of a three-dimensional sample changes the prefactor and leaves the exponent untouched; that immunity licensed the thick-sample geometry of Ref. [1], and made the nanoribbon actively harmful there. Here we seek a gap. A gap is not scale invariant: its position in transverse channel m is set by the channel’s own longitudinal wavevector, $q_m^2 = (\omega^2 - v^2 k_m^2)/v^2$, so a channel sum migrates the

gap frequency channel by channel and smears a Cantor set of gaps into a featureless continuum. The very averaging that protects a power law destroys a gap. Transverse quantisation is therefore not a preference but a prerequisite, in its strictest form: single-mode operation, $N_{\text{prop}} = 1$. That admits only the one-dimensional nanoribbon — the geometry Ref. [1] discarded — and hands us the edge-decoherence problem it avoided. Section 2 is that confrontation. The two papers do not disagree: they report one physical fact, that channel averaging preserves exponents and destroys gaps, read under opposite objectives.

Why Fibonacci, and not the primes.—A gap alone is not topological. The topological content of a one-dimensional quasiperiodic chain rests on the gap-labelling theorem [4], which applies to sequences with a pure-point diffraction spectrum and a cut-and-project construction: Fibonacci [5, 6], Sturmian words, the quadratic irrationals. Under cut-and-project the chain is a slice of a two-dimensional periodic parent, the slice offset ϕ is a synthetic second dimension [7], and each gap carries an integer label q that is the Chern number of that parent. The primes possess none of this structure: their diffraction is closer to limit-periodic than to quasiperiodic, with no phason, no synthetic dimension, and no gap label. A prime-indexed modulation would open a dense family of weak gaps carrying no invariant. We note this explicitly because the arithmetic appeal of such a construction is considerable and its topological content is nil.

If the gaps of a Fibonacci isotope ribbon are genuinely labelled, their positions are fixed by the sequence rather than by any realisation of disorder, and edge branches must traverse gap q exactly q times per phason cycle [7, 8]. Section 3 tests both statements. Section 2 first establishes whether such a ribbon can exist at all.

2 The survival window

2.1 Transfer matrix and localisation length

For a chain of masses M_n with nearest-neighbour force constant K and spacing a ,

$$\begin{pmatrix} u_{n+1} \\ u_n \end{pmatrix} = T_n \begin{pmatrix} u_n \\ u_{n-1} \end{pmatrix}, \quad T_n = \begin{pmatrix} 2 - M_n \omega^2 / K & -1 \\ 1 & 0 \end{pmatrix}, \quad (1)$$

the inverse localisation length is the Lyapunov exponent $\gamma(\omega) = \lim_{N \rightarrow \infty} N^{-1} \ln \|\prod_n T_n\|$, with $\xi = a/\gamma$ and in-gap transmission $\mathcal{T}(L) \sim e^{-2L/\xi}$ [9]. Force constants are isotope-independent within the Born–Oppenheimer approximation, so a single primitive-cell calculation serves every arrangement; only the diagonal mass matrix is rebuilt.

2.2 Gap strength from the mass contrast

The ^{10}BN and ^{11}BN formula units (24.01 vs. 25.01 amu) give a density contrast $\Delta\rho/\rho = 4.1\%$, an impedance contrast $\Delta Z/Z = 2.04\%$, and single-interface reflection $|r| = 0.0102$ [10]. Weak-reflection theory predicts

$$\xi_{\text{gap}} = \frac{d}{2|r|}, \quad \frac{\Delta\omega}{\omega} = \frac{2|r|}{\pi} = 0.67\%, \quad (2)$$

with d the modulation period; direct Lyapunov calculation on 2×10^5 -site chains reproduces both to 1–6%.

2.3 The cost of topology

Quasiperiodicity distributes reflection strength across a Cantor set of gaps instead of one Bragg resonance [5, 11]. A same-frequency comparison settles its price: a periodic superlattice with its Bragg gap at 0.25 THz ($d_{\text{per}} = 40.1$ nm) gives $\xi_{\text{per}} = 1.96 \mu\text{m}$, matching the analytic $d/2|r|$ to three digits, while the Fibonacci (0, 1) gap placed at the same frequency gives $\xi_{\text{gap}} = 4.31 \mu\text{m}$. Topology costs

$$\xi_{\text{fib}}/\xi_{\text{per}} = 2.20 \quad (3)$$

at matched frequency and mass contrast.

2.4 Two perturbations, opposite symmetries

The isotope modulation spans the full ribbon cross-section and is strictly diagonal in the transverse basis, $V_{mm'}^{\text{iso}} = -\omega^2 \epsilon(z) \delta_{mm'}$: it opens a gap in every channel independently, at a channel-dependent frequency — the smearing mechanism restated. Edge roughness breaks transverse translation and is off-diagonal; its consequence depends qualitatively on the mode count. In multi-mode operation it scatters between subbands — genuine decoherence. In single-mode operation there is nowhere to scatter but backwards, and the mechanism reduces to one-dimensional Anderson localisation, which *adds* to γ rather than destroying the gap. A single-mode ribbon converts the dominant loss into one of a different kind; establishing strict single-mode operation is accordingly the central technical requirement.

2.5 The tension and the working point

Single-mode operation with free (Neumann) edges requires $f < v/2W$. Coherence requires $\xi_{\text{dis}} > 3\xi_{\text{gap}}$. With $\xi_{\text{gap}} = C_\xi \lambda$ and the calibrated law of Sec. 2.6, the two combine into

$$\lambda^3 > 24\pi^2 C_\xi S_w(2q) \Rightarrow \lambda \gtrsim 54 [\Delta_r/\text{nm}]^{2/3} \text{ nm}, \quad (4)$$

a far softer wall than the linear bound the Ziman form would imply; the change of exponent from Δ_r to $\Delta_r^{2/3}$ follows from the W^2 law below.

The gaps of a Fibonacci chain sit on the Fourier module,

$$f_{(p,q)} = \frac{v}{2d} |p + q\tau^{-1}|, \quad (5)$$

$\tau = (1 + \sqrt{5})/2$. Placing the $q = 1$ gap at the target frequency fixes $d = \tau^{-1}v/2f = 24.8$ nm (99 lattice sites per letter); a module-targeted scan confirms all four principal gaps within 0.3% of Eq. (5), with labels verified through the integrated density of states (IDOS). Two findings deserve record. The (1, 0) gap — the only one periodic intuition would seek, at $v/2d$ — is thirty times weaker than (0, 1) at this decoration and carries $q = 0$: not topological, and fortunately nearly absent. The (1, −1) gap at 0.154 THz is as strong as (0, 1) ($\xi = 4.18 \mu\text{m}$) with the opposite label; both lie below the single-mode ceiling of a 30 nm ribbon, so one device hosts two topological gaps of opposite Chern number — a built-in control for Sec. 4.

2.6 Calibration: breakdown of the ray formula

The solver is validated three ways before any scan: it reproduces the clean-ribbon limit exactly ($\mathcal{T} = N_{\text{prop}}$ to five digits at three grid spacings, every transmission eigenvalue equal to unity); it passes a realization-matched grid-convergence gate (1.3% between $h = 1$ and 0.5 nm); and it

reproduces the analytic single-bump Born reflection $|r|^2 = (q^2/W^2)|\tilde{a}(2q)|^2$, converging with the grid to a uniform +3.7% excess — a genuine second-order correction whose role below is decisive.

The definitive scan — fifteen single-mode-locked parameter points, 24 disorder seeds and six lengths each, every length window spanning $(0.4\text{--}3)\xi$ of the law itself so that the decay is sampled rather than extrapolated — gives

$$\begin{aligned}\frac{d \ln \xi}{d \ln \Delta_r} &= -1.91 \pm 0.06, & \frac{d \ln \xi}{d \ln W} &= +1.87 \pm 0.08, \\ \frac{d \ln \xi}{d \ln f} &= -1.384 \pm 0.095,\end{aligned}\tag{6}$$

against the law's $(-2, +2, -2+4q^2L_c^2 = -1.38)$: all three within tolerance, the frequency exponent to one part in a thousand. The absolute magnitude matches with median $\xi_{\text{meas}}/\xi_{\text{law}} = 0.965$, while the single-bump test independently predicts $1/1.037 = 0.964$ for the same second-order excess: two measurements of one correction, one deterministic and one ensemble-averaged, agreeing to 10^{-3} . Across the scan the Ziman formula underestimates ξ by a median factor of 10.8: in the single-mode wave limit $\lambda \gg W$ a rough edge is Born-transparent, and the ray-limit picture of diffuse wall collisions does not apply. The scaling structure — the rate $q^2 S_w(2q)/W^2$ — agrees with the perturbative treatment of roughness scattering in few-mode elastic waveguides given by Santamore and Cross [12] in the context of the thermal-conductance quantum, there for clamped elastic boundaries; the closed zero-parameter form of Eq. (??), its absolute normalization, and its verification deep into the localized regime are the additions made here. The residuals retain a mild trend in qL_c (from 0.78 at $qL_c = 0.16$ to 1.12 at 0.42), the expected fingerprint of second-order structure beyond the leading law.

Methodological record — three failure modes, each caught by a different gate and none by inspection. A staircase-masked solver produced grid-divergent spurious *excess* scattering ($\xi = 120/42/16$ nm at $h = 2/1/0.5$ nm; no continuum limit), caught by grid-convergence testing. A naive finite-difference discretization of the transformed operator was non-Hermitian and current-violating (transmission eigenvalues up to 1.26) while remaining grid-*converged* and Born-consistent in total transmission: invisible to convergence testing, exposed only by a unitarity gate on the transmission-eigenvalue spectrum, and cured by assembling the discrete operator from the quadratic form of the problem, which makes Hermiticity ($R_{m+1} = P_m^T$ by construction) and the exact tilted-wall Neumann condition properties of the construction rather than of a hand-derived stencil. Length windows tied to the Ziman estimate — eleven times too short — sample only shallow decay and biased the W exponent by +0.3 in a preliminary pass: windows must track the law being tested. And a scan whose ladders crossed a transverse-mode threshold returned exponents of opposite sign on one axis (+5.45 and -2.91), the diagnostic signature of an unenforced single-mode condition.

Table 1: Working point. ξ_{dis} from the calibrated law of Sec. 2.6; margins are $\xi_{\text{dis}}/\xi_{\text{gap}}$.

frequency / operating gap	0.250 THz, (0, 1), $q = 1$
ribbon width W / period d	30 nm / 24.8 nm
ξ_{gap}	4.31 μm
ξ_{dis} at $\Delta_r = 0.6$ (1.0) nm	76 (30) μm
coherence margin	18 $\times$ (7.0 $\times$)

The consequence is the central practical result of this section: coherence no longer demands $\Delta_r \lesssim 0.13$ nm — a single atomic row — but survives $\Delta_r = 0.6$ nm with an eighteen-fold margin and clears the $3\times$ threshold at 1.0 nm (Table 1, Fig. ??): within routine electron-beam lithography.

A scope statement, owed to the reader before the topology of Sec. 3 is built on these numbers: everything above treats a single scalar polarization with linear dispersion $v = 20$ nm/ps. A real suspended h-BN ribbon carries four low-frequency branches — longitudinal, in-plane transverse, torsional, and a flexural branch whose dispersion is quadratic and which dominates the low-frequency density of states. The two mechanisms this paper rests on are polarization-generic: gap labelling requires only one-dimensional quasiperiodicity of the mass sequence, and Born transparency only q -selective backscattering from a weak boundary perturbation, so the structure of Eq. (??) and of the module (5) carries over branch by branch. The numbers do not: mode mixing among branches, the quadratic flexural dispersion, and branch-dependent edge coupling will shift working-point frequencies, ξ values and margins, and the scalar model is best read as a quantitative treatment of the longitudinal channel and a structural template for the rest. A full elastic calculation is the natural next step and lies outside this paper.

3 Topological verification

3.1 Gap labelling, falsifiably

Within each gap the IDOS is constant and restricted by the gap-labelling theorem to $\mathcal{N} = (p + q\tau^{-1}) \bmod 1$ [4]. We compute $\mathcal{N}$ by Sturm node counting on the transfer-matrix recursion ($\mathcal{O}(N)$, no diagonalisation); Table 2 lists the twelve strongest gaps of a 6×10^4 -site chain (Fig. 1). The two strongest take the textbook values τ^{-1} and $1 - \tau^{-1}$.

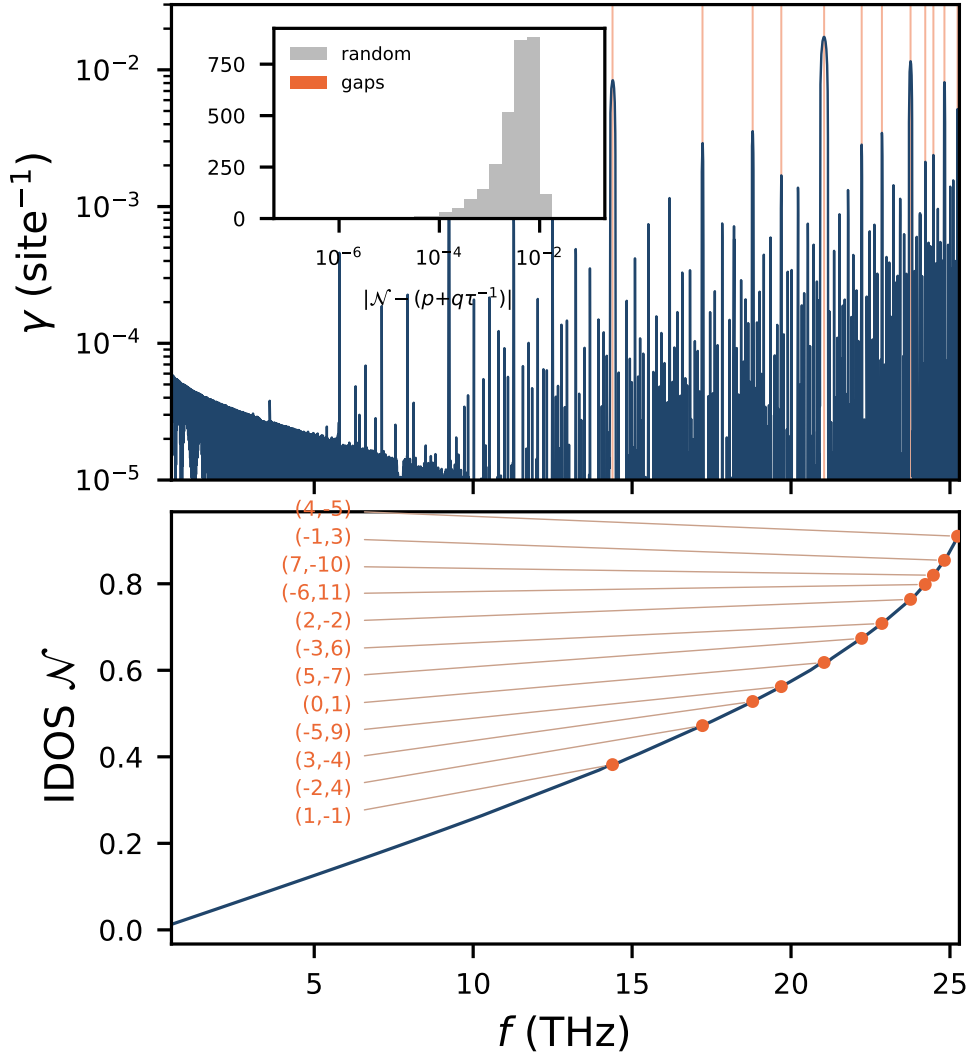


Figure 1: Lyapunov spectrum $\gamma(\omega)$ (top) and IDOS staircase (bottom) of the Fibonacci chain, with the twelve labelled gaps of Table 2. Inset: gap-labelling residuals, measured gaps against 3×10^3 random controls (log axis) — the falsifiability argument of Sec. 3 in one picture.

Taken at face value the test is vacuous: $\mathbb{Z} + \tau^{-1}\mathbb{Z}$ is dense in $[0, 1]$, so some pair fits any number. The content is quantitative. Against 5×10^3 uniform controls under the identical fitting procedure ($|q| \leq 30$): measured gaps show median residual 2.95×10^{-6} (maximum 5.95×10^{-6}); random controls, 4.08×10^{-3} — a separation of 1385; the probability that a random value fits as well as the *worst* measured gap is $p = 1.4 \times 10^{-3}$. We recommend this control whenever gap labelling is claimed numerically.

3.2 Phason pumping: the full counter-propagating structure

The phason angle ϕ of the cut-and-project word $s_n(\phi) = \lfloor (n+2)\tau^{-1} + \phi \rfloor - \lfloor (n+1)\tau^{-1} + \phi \rfloor$ is the synthetic dimension; a full cycle traverses the hull, and edge branches must cross gap q exactly q times [7]. Tracking such branches requires care: sorting by frequency exchanges identities at band

Table 2: The twelve principal gaps ($N = 6 \times 10^4$ sites). ξ at gap centre; residual is $|\mathcal{N} - (p + q\tau^{-1})|$ for the best pair with $|q| \leq 30$.

f (THz)	γ (site $^{-1}$)	ξ (nm)	IDOS	(p, q)	resid.
14.386	0.00839	29.9	0.38197	(1, -1)	7×10^{-7}
17.213	0.00290	86.5	0.47213	(-2, 4)	3×10^{-6}
18.792	0.00356	70.3	0.52787	(3, -4)	3×10^{-6}
19.693	0.00170	147.4	0.56230	(-5, 9)	6×10^{-6}
21.040	0.01744	14.4	0.61803	(0, 1)	7×10^{-7}
22.222	0.00284	88.1	0.67377	(5, -7)	5×10^{-6}
22.859	0.00345	72.6	0.70820	(-3, 6)	4×10^{-6}
23.760	0.01155	21.7	0.76393	(2, -2)	1×10^{-6}
24.222	0.00185	135.1	0.79837	(-6, 11)	7×10^{-6}
24.479	0.00239	104.7	0.81967	(7, -10)	7×10^{-6}
24.826	0.00812	30.8	0.85410	(-1, 3)	2×10^{-6}
25.239	0.00516	48.5	0.90983	(4, -5)	3×10^{-6}

crossings, so we continue states by eigenvector overlap with Hungarian assignment, terminating any branch whose best overlap falls below 0.15, and record left- and right-end weights separately (Fig. 2).

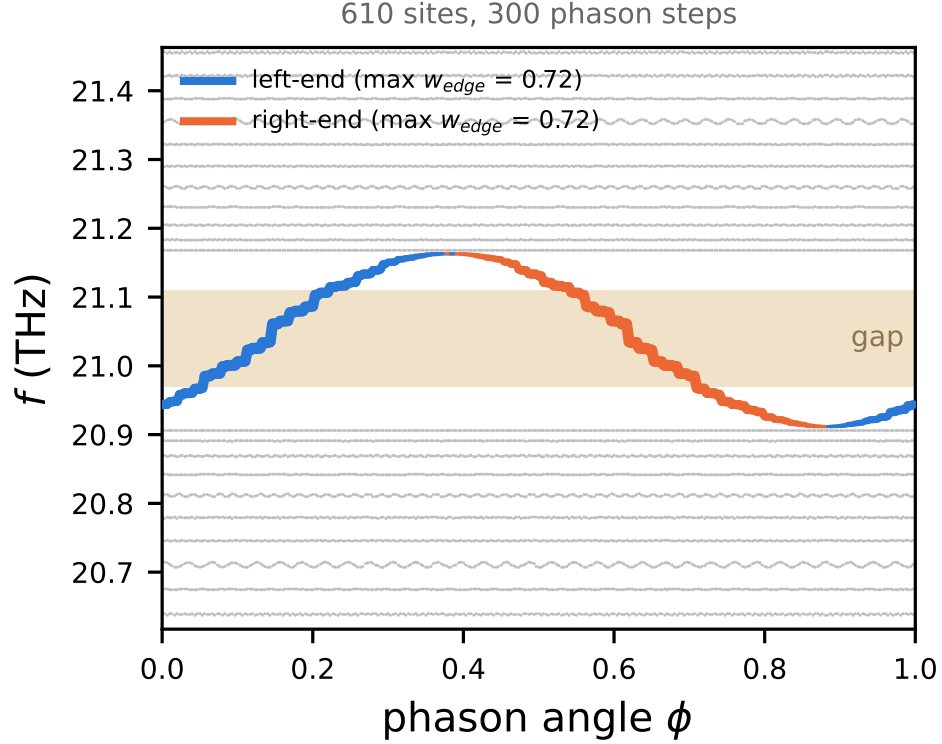


Figure 2: Phason spectrum of the $(0,1)$ gap: frequency against ϕ , bulk states as points, edge-bound branches coloured by the *instantaneous* dominant end (blue left, orange right), line width $\propto$ edge weight. The colour change at the in-gap degeneracy is the identity exchange of Sec. 3C.

A first pass on the $(0,1)$ gap (610-site free chain, 400 steps) tracked 27 branches and reported a single edge-bound branch — right end, weight 0.721, sweeping 86% of the gap with one kink — stable across chain lengths 610–1597 and steps 400–800. Taken alone this suggested a one-sided pump. Two basis-independent measurements correct that reading. *Spectral flow*: counting eigenvalues below each gap centre at 600 phason steps, every one of the twelve gaps shows exactly $|q|$ downward and $|q|$ upward crossings per cycle — 1+1 for $(0,1)$, 3+3 for $(-1,3)$, up to 11+11 for $(-6,11)$ — without exception. *Crossing-moment classification*: diagonalizing with eigenvectors at each detected crossing, the descending crossings of every $q > 0$ gap are right-bound and the ascending left-bound, exactly mirrored for $q < 0$; for $(-1,3)$ all six crossings alternate side-perfectly, and the $(0,1)$ pair carries weights 0.72 and 0.71. The counter-propagating pair exists in full; the per-edge spectral flow equals $\pm q$; the bulk–boundary correspondence holds in its standard form for all twelve labels.

3.3 A cautionary note on overlap tracking

Why did the first pass see one branch? The two counter-propagating edge modes hybridize where they are degenerate inside the gap; at that avoided crossing, overlap continuation — by construction — follows the smoothly deforming hybrids and exchanges identities, stitching the descending halves of the two branches into a single apparent branch with one kink: precisely the phenomenology reported. The spectral-flow classification never asks for continuity between steps

and is immune. Under fixed (Dirichlet) ends the pair persists (crossings at $\phi = 0.833, 0.932$ against $0.618, 0.147$ free), so termination relocates the crossings but removes nothing. We record the artefact because Hungarian-overlap continuation is standard practice for pump spectra, and it fails silently in exactly the situation — counter-propagating branches meeting inside a gap — that topological pumps guarantee.

4 Experimental design

4.1 Frequency band; exclusion of transient gratings

The working gap sits at 0.250 THz, $\lambda = 80$ nm. The natural wavevector-selective probe, the transient thermal grating, is excluded by optics: two beams of wavelength λ_{opt} cannot write a grating finer than $\lambda_{\text{opt}}/2$, so a 400 nm pump bottoms out near 200 nm — $2.5\times$ too coarse — and 266 nm reaches only 133 nm. No geometry closes the factor. The measurement must be performed in the time domain: asynchronous optical sampling (ASOPS) between two femtosecond oscillators offset by a few kHz [13] provides ~ 100 fs effective resolution across the full 12.5 ns repetition window, sampling a 4 ps acoustic period over $\sim 3 \times 10^3$ cycles — spectral resolution ~ 0.3 GHz.

4.2 Cavity-mode spectroscopy of the gap

Aluminium pads (~ 20 nm) on both ends serve as broadband phonon sources [14], detection mirrors, and the reflectors of an acoustic Fabry–Pérot cavity; the ribbon interior is interrogated through its mode spectrum alone. A cavity of $L = 20 \mu\text{m}$ has free spectral range $v/2L = 0.5$ GHz; the $(0, 1)$ gap (1.7 GHz) expels three to four consecutive modes, and with $\xi_{\text{gap}} = 4.31 \mu\text{m}$ the in-gap transmission $e^{-2L/\xi} \approx 10^{-4}$: modes are removed, not shifted. The observable is differential, as in Ref. [1]: the Fibonacci ribbon is paired on-chip with a composition-matched random permutation of the same isotope content; boundary, transducer and anharmonic contributions cancel to first order (Fig. 3). The resolution budget is stated plainly: five elements across the $(0, 1)$ gap, three across $(1, -1)$ — sufficient to establish absence of modes, insufficient for lineshape work.

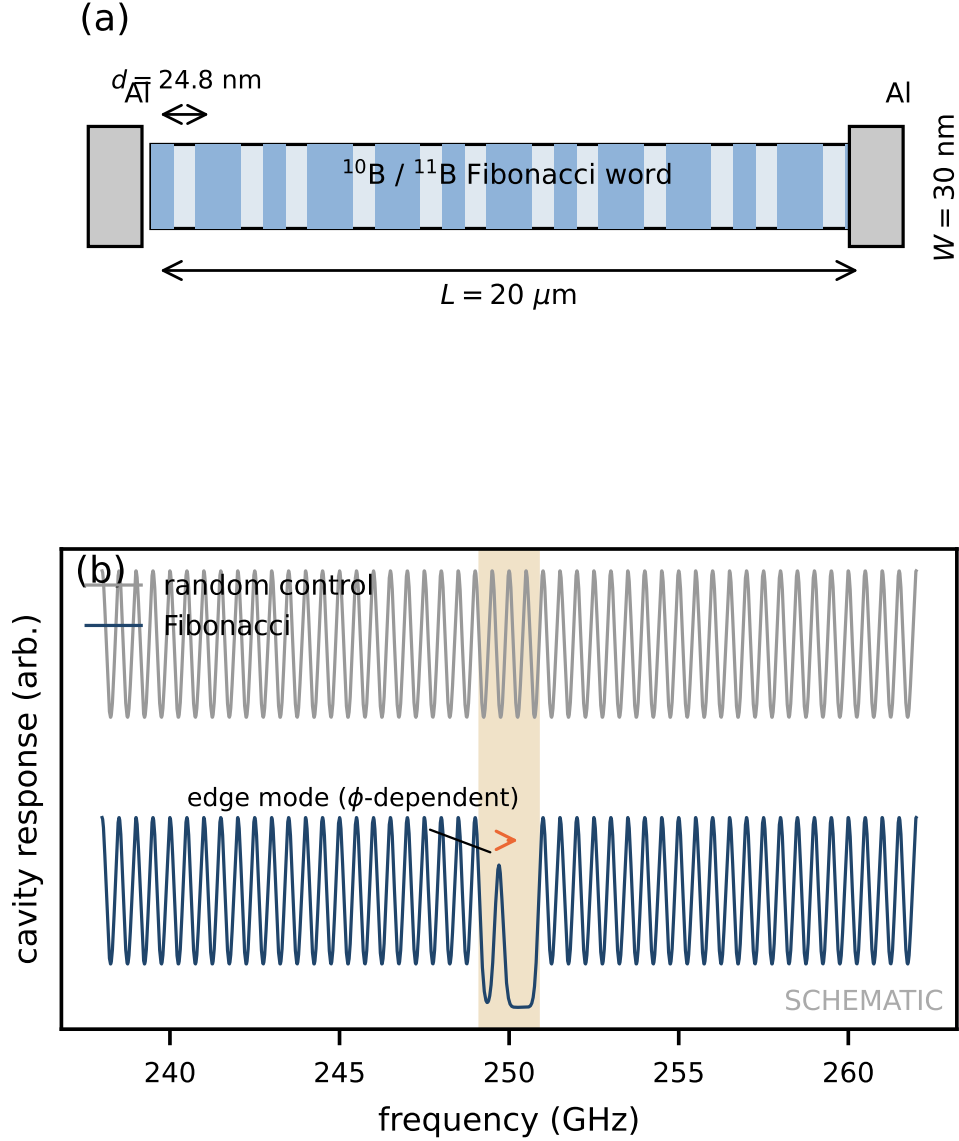


Figure 3: Proposed measurement (schematic). (a) Ribbon with Fibonacci isotope letters and Al transducers. (b) Illustrative cavity spectra: composition-matched random control (grey, full mode comb) against Fibonacci (blue, gap emptied), with the ϕ -dependent edge mode inside the gap.

4.3 Edge-state pumping by a phason array

The phason angle is fixed lithographically, one value per ribbon; an array of ~ 20 ribbons at $\phi = 0, 0.05, \dots, 0.95$ converts the pump cycle into chip position. Section 3.2 predicts an in-gap resonance whose frequency advances monotonically across the array, present for most of the cycle. Two structural controls come free. The $(1, -1)$ gap in the same ribbons carries the opposite label, so its in-gap resonance must wind oppositely across the same array; any trivial systematic — strain, thickness drift, composition gradient — moves both gaps identically and is excluded. And the counter-propagating structure is directly testable: the descending resonance ($\phi \approx 0.62$ for

(0,1)) lives at one end and the ascending one ($\phi \approx 0.15$) at the other, so exchanging pump and probe transducers must swap which portion of the array shows the stronger in-gap response — a signature no single-sided mechanism reproduces.

4.4 Sample specification

$W = 30$ nm (single-mode to 0.267 THz); $L = 20$ μm ($4.6 \xi_{\text{gap}}$); $d = 24.8$ nm (99 sites per letter); ~ 800 letters per ribbon ($F_{16} = 987$ truncated); $\Delta_r \leq 0.6$ nm rms ($18\times$ margin); $T \leq 4$ K. The isotope pattern is one-dimensional and in-plane, within electron-beam masking resolution, with sequential two-precursor regrowth as the candidate route. The contrast with Ref. [1] bears repeating: that design required 2.4×10^5 independently specified units over 300 μm ; this one, ~ 800 letters over 20 μm at mesoscopic pitch, with edge tolerances relaxed an order of magnitude by the physics of Sec. 2.6. The tension that forced the working point down in frequency also carried it into fabricability.

Acknowledgments Numerical calculations were performed with open-source software. All code, calibration data, and figure-generation scripts sufficient to reproduce every number in this manuscript are available at <https://github.com/Ruqing1963/hbn-fibonacci-topological-phonons>.

References

- [1] Ronghua Wu and Ruqing Chen. Hyperuniform isotope arrangements and the scaling of phonon scattering in hexagonal boron nitride. Zenodo preprint, 2026. <https://zenodo.org/records/21981511>.
- [2] Salvatore Torquato and Frank H. Stillinger. Local density fluctuations, hyperuniformity, and order metrics. *Phys. Rev. E*, 68:041113, 2003.
- [3] Shin-ichiro Tamura. Isotope scattering of dispersive phonons in Ge. *Phys. Rev. B*, 27:858, 1983.
- [4] Jean Bellissard, Anton Bovier, and Jean-Michel Ghez. Gap labelling theorems for one dimensional discrete Schrödinger operators. *Rev. Math. Phys.*, 4:1, 1992.
- [5] Mahito Kohmoto, Leo P. Kadanoff, and Chao Tang. Localization problem in one dimension: mapping and escape. *Phys. Rev. Lett.*, 50:1870, 1983.
- [6] Stellan Ostlund, Rahul Pandit, David Rand, Hans Joachim Schellnhuber, and Eric D. Siggia. One-dimensional Schrödinger equation with an almost periodic potential. *Phys. Rev. Lett.*, 50:1873, 1983.
- [7] Yaacov E. Kraus, Yoav Lahini, Zohar Ringel, Mor Verbin, and Oded Zilberberg. Topological states and adiabatic pumping in quasicrystals. *Phys. Rev. Lett.*, 109:106402, 2012.
- [8] David J. Apigo, Wenting Cheng, Kyle F. Dobiszewski, Emil Prodan, and Camelia Prodan. Observation of topological edge modes in a quasiperiodic acoustic waveguide. *Phys. Rev. Lett.*, 122:095501, 2019.
- [9] P. W. Anderson. Absence of diffusion in certain random lattices. *Phys. Rev.*, 109:1492, 1958.

- [10] T. Q. P. Vuong, S. Liu, A. Van der Lee, R. Cuscó, L. Artús, T. Michel, P. Valvin, J. H. Edgar, G. Cassabois, and B. Gil. Isotope engineering of van der Waals interactions in hexagonal boron nitride. *Nat. Mater.*, 17:152, 2018.
- [11] R. Merlin, K. Bajema, Roy Clarke, F.-Y. Juang, and P. K. Bhattacharya. Quasiperiodic GaAs-AlAs heterostructures. *Phys. Rev. Lett.*, 55:1768, 1985.
- [12] D. H. Santamore and M. C. Cross. Effect of surface roughness on the universal thermal conductance. *Phys. Rev. B*, 63:184306, 2001. VERIFY against original before submission: perturbative roughness scattering in few-mode elastic waveguides.
- [13] A. Bartels, R. Cerna, C. Kistner, A. Thoma, F. Hudert, C. Janke, and T. Dekorsy. Ultrafast time-domain spectroscopy based on high-speed asynchronous optical sampling. *Rev. Sci. Instrum.*, 78:035107, 2007.
- [14] C. Thomsen, H. T. Grahn, H. J. Maris, and J. Tauc. Surface generation and detection of phonons by picosecond light pulses. *Phys. Rev. B*, 34:4129, 1986.